

Double Robustness

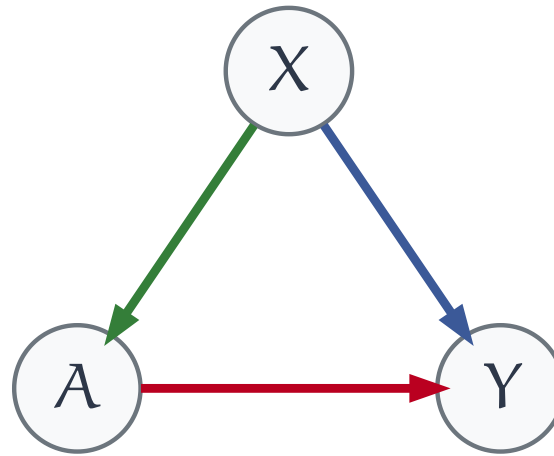
CML Week 4

2026-09-22

Schedule

1. Design-based causal inference & Monte Carlo simulation
2. Covariate adjustment
3. Balancing weights
4. **Double robustness, Double ML and TMLE**
5. Heterogeneous Treatment Effects
6. HTE II: Neural Networks
7. Off-policy evaluation and optimization
8. Experimental design
9. Panel data and the new diff-in-diff
10. Partial Identification
11. Response adaptation
12. Interference

The big idea



- › The edge from $A \rightarrow Y$ is the causal effect we want to measure
- › The edge from $X \rightarrow Y$ is what we're correcting for when we do covariate adjustment
- › The edge from $X \rightarrow A$ is what we're correcting for when we do balancing
- › What if we simultaneously do **both**? How can we do that?



(Bang and Robins 2005)

Double robustness

The core insight: An estimator is doubly robust if it remains consistent when *either* (but not necessarily both):

1. **Outcome regression model** is correctly specified, *or*
2. **Treatment assignment model** (propensity score) is correctly specified

Why this matters:

- › Model misspecification is inevitable in practice
- › We get “two chances” to be right instead of one
- › Combines the best features of:
 - » **G-computation:** models the outcome regression $\mu(X) = E[Y | A = 1, X]$
 - » **Inverse probability weighting:** models the propensity score $e(X) = P(A = 1 | X)$ and reweights

Key theoretical result: Under standard assumptions, DR estimators achieve the semiparametric efficiency bound when both models are correct.



Setup and notation

- › **Data:** $(A_i, X_i, Y_i)_{i=1}^n$, with $A_i \in \{0, 1\}$ a sampling indicator: only see Y_i when $A_i = 1$
- › **Target:** $\psi = E[Y]$
- › **Identification:** MAR, $Y \perp A \mid X$

Two nuisance functions, in last week's notation, each with a parametric working model:

	Truth	Working model
Propensity score	$e(X) = P(A = 1 \mid X)$	$e_\alpha(X)$
Outcome regression	$\mu(X) = E[Y \mid A = 1, X]$	$\mu_{\beta, \phi}(X)$

Bang & Robins write $\pi(X; \alpha)$, $e(X; \beta, \phi)$ and μ . We write $e_\alpha(X)$, $\mu_{\beta, \phi}(X)$ and ψ .



The DR estimator

Bang & Robins, linear case. Three steps:

1. **Fit the propensity model:** $\hat{e}(X) = e_{\hat{\alpha}}(X)$

The DR estimator

Bang & Robins, linear case. Three steps:

1. **Fit the propensity model:** $\hat{e}(X) = e_{\hat{\alpha}}(X)$
2. **Fit the outcome model with $\hat{e}^{-1}$ as an extra regressor**, by OLS among the observed ($A_i = 1$):

$$\mu_{\beta, \phi}(X_i) = X_i^\top \beta + \phi \hat{e}(X_i)^{-1}, \quad 0 = \sum_i A_i \begin{pmatrix} X_i \\ \hat{e}(X_i)^{-1} \end{pmatrix} (Y_i - \mu_{\beta, \phi}(X_i))$$

Call the fit $\hat{\mu} = \mu_{\hat{\beta}, \hat{\phi}}$.



The DR estimator

Bang & Robins, linear case. Three steps:

1. **Fit the propensity model:** $\hat{e}(X) = e_{\hat{\alpha}}(X)$
2. **Fit the outcome model with $\hat{e}^{-1}$ as an extra regressor**, by OLS among the observed ($A_i = 1$):

$$\mu_{\beta, \phi}(X_i) = X_i^\top \beta + \phi \hat{e}(X_i)^{-1}, \quad 0 = \sum_i A_i \begin{pmatrix} X_i \\ \hat{e}(X_i)^{-1} \end{pmatrix} (Y_i - \mu_{\beta, \phi}(X_i))$$

Call the fit $\hat{\mu} = \mu_{\hat{\beta}, \hat{\phi}}$.

3. **Average the fitted values over everyone**, observed or not:

$$\hat{\psi}_{\text{dr}} = \frac{1}{n} \sum_i \hat{\mu}(X_i)$$



The DR estimator

Bang & Robins, linear case. Three steps:

1. **Fit the propensity model:** $\hat{e}(X) = e_{\hat{\alpha}}(X)$
2. **Fit the outcome model with $\hat{e}^{-1}$ as an extra regressor**, by OLS among the observed ($A_i = 1$):

$$\mu_{\beta, \phi}(X_i) = X_i^\top \beta + \phi \hat{e}(X_i)^{-1}, \quad 0 = \sum_i A_i \begin{pmatrix} X_i \\ \hat{e}(X_i)^{-1} \end{pmatrix} (Y_i - \mu_{\beta, \phi}(X_i))$$

Call the fit $\hat{\mu} = \mu_{\hat{\beta}, \hat{\phi}}$.

3. **Average the fitted values over everyone**, observed or not:

$$\hat{\psi}_{\text{dr}} = \frac{1}{n} \sum_i \hat{\mu}(X_i)$$

Step 3 is just averaging over the model. All we did was add $\hat{e}^{-1}$ in step 2.



Why is it DR? Step 0: rewrite the estimator

Shorthand: $\hat{e}_i = \hat{e}(X_i)$ and $\hat{\mu}_i = \hat{\mu}(X_i)$.

Truth: $e(X) = P(A = 1 \mid X)$, $\mu(X) = E[Y \mid A = 1, X]$, and $\psi = E[Y] = E[\mu(X)]$.

› $\hat{e}_i^{-1}$ is a **regressor** in the outcome fit, its normal equation holds in sample:

$$\sum_i A_i \hat{e}_i^{-1} (Y_i - \hat{\mu}_i) = 0$$



Why is it DR? Step 0: rewrite the estimator

Shorthand: $\hat{e}_i = \hat{e}(X_i)$ and $\hat{\mu}_i = \hat{\mu}(X_i)$.

Truth: $e(X) = P(A = 1 | X)$, $\mu(X) = E[Y | A = 1, X]$, and $\psi = E[Y] = E[\mu(X)]$.

› $\hat{e}_i^{-1}$ is a **regressor** in the outcome fit, its normal equation holds in sample:

$$\sum_i A_i \hat{e}_i^{-1} (Y_i - \hat{\mu}_i) = 0$$

› Add this zero to $\hat{\psi}_{\text{dr}}$ and it becomes a sum of two familiar pieces:

$$\hat{\psi}_{\text{dr}} = \underbrace{\frac{1}{n} \sum_i \hat{\mu}_i}_{\text{G-computation}} + \underbrace{\frac{1}{n} \sum_i \frac{A_i}{\hat{e}_i} (Y_i - \hat{\mu}_i)}_{\text{IPW, applied to residuals}}$$



Why is it DR? Step 0: rewrite the estimator

Shorthand: $\hat{e}_i = \hat{e}(X_i)$ and $\hat{\mu}_i = \hat{\mu}(X_i)$.

Truth: $e(X) = P(A = 1 | X)$, $\mu(X) = E[Y | A = 1, X]$, and $\psi = E[Y] = E[\mu(X)]$.

- › $\hat{e}_i^{-1}$ is a **regressor** in the outcome fit, its normal equation holds in sample:

$$\sum_i A_i \hat{e}_i^{-1} (Y_i - \hat{\mu}_i) = 0$$

- › Add this zero to $\hat{\psi}_{\text{dr}}$ and it becomes a sum of two familiar pieces:

$$\hat{\psi}_{\text{dr}} = \underbrace{\frac{1}{n} \sum_i \hat{\mu}_i}_{\text{G-computation}} + \underbrace{\frac{1}{n} \sum_i \frac{A_i}{\hat{e}_i} (Y_i - \hat{\mu}_i)}_{\text{IPW, applied to residuals}}$$

- › This is the **augmented IPW** form. Plan: hold one model at its truth, let the other be anything, and take expectations term by term using **MAR** ($Y \perp A | X$) and **iterated expectations**.



Why is it DR? Step 1: the outcome model is right

Suppose $\hat{\mu}_i \rightarrow \mu(X_i)$. Make no assumption about $\hat{e}$ beyond $\hat{e} > 0$.

First term. MAR lets us drop the conditioning on $A = 1$:

$$E[\mu(X)] = E[E[Y | A = 1, X]] = E[E[Y | X]] = E[Y] = \psi$$

Why is it DR? Step 1: the outcome model is right

Suppose $\hat{\mu}_i \rightarrow \mu(X_i)$. Make no assumption about $\hat{e}$ beyond $\hat{e} > 0$.

First term. MAR lets us drop the conditioning on $A = 1$:

$$E[\mu(X)] = E[E[Y | A = 1, X]] = E[E[Y | X]] = E[Y] = \psi$$

Second term. Condition on X ; $\hat{e}(X)$ is then a constant and comes outside:

$$E\left[\frac{A}{\hat{e}(X)} (Y - \mu(X))\right] = E\left[\frac{1}{\hat{e}(X)} E[A (Y - \mu(X)) | X]\right]$$

and the inner expectation is zero because $A(Y - \mu)$ is only nonzero when $A = 1$, where μ is the conditional mean:

$$E[A (Y - \mu(X)) | X] = e(X) E[Y - \mu(X) | A = 1, X] = e(X) \cdot 0 = 0$$



Why is it DR? Step 1: the outcome model is right

Suppose $\hat{\mu}_i \rightarrow \mu(X_i)$. Make no assumption about $\hat{e}$ beyond $\hat{e} > 0$.

First term. MAR lets us drop the conditioning on $A = 1$:

$$E[\mu(X)] = E[E[Y | A = 1, X]] = E[E[Y | X]] = E[Y] = \psi$$

Second term. Condition on X ; $\hat{e}(X)$ is then a constant and comes outside:

$$E\left[\frac{A}{\hat{e}(X)} (Y - \mu(X))\right] = E\left[\frac{1}{\hat{e}(X)} E[A (Y - \mu(X)) | X]\right]$$

and the inner expectation is zero because $A(Y - \mu)$ is only nonzero when $A = 1$, where μ is the conditional mean:

$$E[A (Y - \mu(X)) | X] = e(X) E[Y - \mu(X) | A = 1, X] = e(X) \cdot 0 = 0$$

- › The correction term vanishes for **any** $\hat{e}$: a wrong propensity model just reweights residuals that are already mean-zero given X . So $\hat{\psi}_{\text{dr}} \rightarrow \psi$; the correction only adds noise.



Why is it DR? Step 2: the propensity score is right

Suppose $\hat{e}_i \rightarrow e(X_i)$, and let $\hat{\mu}_i \rightarrow \mu^*(X_i)$ for an arbitrary (wrong) μ^* .

Regroup the two pieces by what multiplies μ^* :

$$\hat{\psi}_{\text{dr}} = \frac{1}{n} \sum_i \frac{A_i Y_i}{e(X_i)} + \frac{1}{n} \sum_i \left(1 - \frac{A_i}{e(X_i)} \right) \mu^*(X_i)$$



Why is it DR? Step 2: the propensity score is right

Suppose $\hat{e}_i \rightarrow e(X_i)$, and let $\hat{\mu}_i \rightarrow \mu^*(X_i)$ for an arbitrary (wrong) μ^* .

Regroup the two pieces by what multiplies μ^* :

$$\hat{\psi}_{\text{dr}} = \frac{1}{n} \sum_i \frac{A_i Y_i}{e(X_i)} + \frac{1}{n} \sum_i \left(1 - \frac{A_i}{e(X_i)}\right) \mu^*(X_i)$$

First term. Horvitz–Thompson (last week). MAR splits $E[AY \mid X]$ into a product:

$$E \left[\frac{AY}{e(X)} \right] = E \left[\frac{E[A \mid X] E[Y \mid X]}{e(X)} \right] = E[E[Y \mid X]] = \psi$$



Why is it DR? Step 2: the propensity score is right

Suppose $\hat{e}_i \rightarrow e(X_i)$, and let $\hat{\mu}_i \rightarrow \mu^*(X_i)$ for an arbitrary (wrong) μ^* .

Regroup the two pieces by what multiplies μ^* :

$$\hat{\psi}_{\text{dr}} = \frac{1}{n} \sum_i \frac{A_i Y_i}{e(X_i)} + \frac{1}{n} \sum_i \left(1 - \frac{A_i}{e(X_i)}\right) \mu^*(X_i)$$

First term. Horvitz–Thompson (last week). MAR splits $E[AY \mid X]$ into a product:

$$E \left[\frac{AY}{e(X)} \right] = E \left[\frac{E[A \mid X] E[Y \mid X]}{e(X)} \right] = E[E[Y \mid X]] = \psi$$

Second term. Condition on X ; only A is random inside:

$$E \left[\left(1 - \frac{A}{e(X)}\right) \mu^*(X) \right] = E \left[\left(1 - \frac{E[A \mid X]}{e(X)}\right) \mu^*(X) \right] = E[0 \cdot \mu^*(X)] = 0$$



Why is it DR? Step 2: the propensity score is right

Suppose $\hat{e}_i \rightarrow e(X_i)$, and let $\hat{\mu}_i \rightarrow \mu^*(X_i)$ for an arbitrary (wrong) μ^* .

Regroup the two pieces by what multiplies μ^* :

$$\hat{\psi}_{\text{dr}} = \frac{1}{n} \sum_i \frac{A_i Y_i}{e(X_i)} + \frac{1}{n} \sum_i \left(1 - \frac{A_i}{e(X_i)}\right) \mu^*(X_i)$$

First term. Horvitz–Thompson (last week). MAR splits $E[AY \mid X]$ into a product:

$$E \left[\frac{AY}{e(X)} \right] = E \left[\frac{E[A \mid X] E[Y \mid X]}{e(X)} \right] = E[E[Y \mid X]] = \psi$$

Second term. Condition on X ; only A is random inside:

$$E \left[\left(1 - \frac{A}{e(X)}\right) \mu^*(X) \right] = E \left[\left(1 - \frac{E[A \mid X]}{e(X)}\right) \mu^*(X) \right] = E[0 \cdot \mu^*(X)] = 0$$

- › The outcome model enters only through a term with mean **exactly zero**: it cannot bias, only change variance. Steps 1 and 2 are mirror images: when one model is right, the other's term is inert.



Why is it DR? Both at once: the bias is a product

Let both limits be wrong: $\hat{\mu} \rightarrow \mu^*$ and $\hat{e} \rightarrow e^*$. Taking expectations of the augmented form, with $\psi = E[\mu(X)]$:

$$E[\hat{\psi}_{\text{dr}}] - \psi = E[\mu^*(X)] + E\left[\frac{e(X)}{e^*(X)} (\mu(X) - \mu^*(X))\right] - E[\mu(X)]$$



Why is it DR? Both at once: the bias is a product

Let both limits be wrong: $\hat{\mu} \rightarrow \mu^*$ and $\hat{e} \rightarrow e^*$. Taking expectations of the augmented form, with $\psi = E[\mu(X)]$:

$$E[\hat{\psi}_{\text{dr}}] - \psi = E[\mu^*(X)] + E\left[\frac{e(X)}{e^*(X)} (\mu(X) - \mu^*(X))\right] - E[\mu(X)]$$

Collect the $\mu - \mu^*$ terms:

$$E[\hat{\psi}_{\text{dr}}] - \psi = E\left[(\mu(X) - \mu^*(X)) \frac{e(X) - e^*(X)}{e^*(X)}\right]$$

Why is it DR? Both at once: the bias is a product

Let both limits be wrong: $\hat{\mu} \rightarrow \mu^*$ and $\hat{e} \rightarrow e^*$. Taking expectations of the augmented form, with $\psi = E[\mu(X)]$:

$$E[\hat{\psi}_{\text{dr}}] - \psi = E[\mu^*(X)] + E\left[\frac{e(X)}{e^*(X)} (\mu(X) - \mu^*(X))\right] - E[\mu(X)]$$

Collect the $\mu - \mu^*$ terms:

$$E[\hat{\psi}_{\text{dr}}] - \psi = E\left[(\mu(X) - \mu^*(X)) \frac{e(X) - e^*(X)}{e^*(X)}\right]$$

- › **Either factor zero, bias zero.** Steps 1 and 2 in a single line.
- › **Both a little wrong:** the bias is a **product of two errors**, which is smaller than either. This is the seed of double ML: rates multiply.
- › **Both moderately wrong, with small e^* :** the product is not small, and the denominator makes it worse. This is Kang & Schafer's cautionary tale.



OLS is DR? Step 1: Frisch–Waugh–Lovell

Regress Y on A and X (intercept in X); call the coefficient on A $\hat{\tau}_{OLS}$. **Partial regression** purges X from both variables first: $\tilde{A}_i = A_i - X_i^\top \hat{\gamma}$ and $\tilde{Y}_i = Y_i - X_i^\top \hat{\delta}$.



OLS is DR? Step 1: Frisch–Waugh–Lovell

Regress Y on A and X (intercept in X); call the coefficient on A $\hat{\tau}_{OLS}$. **Partial regression** purges X from both variables first: $\tilde{A}_i = A_i - X_i^\top \hat{\gamma}$ and $\tilde{Y}_i = Y_i - X_i^\top \hat{\delta}$. **FWL**: the multiple-regression coefficient is the regression of residual on residual,

$$\hat{\tau}_{OLS} = \frac{\sum_i \tilde{A}_i \tilde{Y}_i}{\sum_i \tilde{A}_i^2}$$

an **exact in-sample identity**.



OLS is DR? Step 1: Frisch–Waugh–Lovell

Regress Y on A and X (intercept in X); call the coefficient on A $\hat{\tau}_{OLS}$. **Partial regression** purges X from both variables first: $\tilde{A}_i = A_i - X_i^\top \hat{\gamma}$ and $\tilde{Y}_i = Y_i - X_i^\top \hat{\delta}$. **FWL**: the multiple-regression coefficient is the regression of residual on residual,

$$\hat{\tau}_{OLS} = \frac{\sum_i \tilde{A}_i \tilde{Y}_i}{\sum_i \tilde{A}_i^2}$$

an **exact in-sample identity**.

Two readings of the same formula:

- › $\hat{\tau}_{OLS} = \sum_i w_i \tilde{Y}_i$ with $w_i = \tilde{A}_i / \sum_j \tilde{A}_j^2$: a **weighting estimator**, and $\sum_i w_i X_i = 0$ is exact mean balance on X (last week's constraint with $f(X) = X$).
- › $X_i^\top \hat{\gamma}$ is a fitted **linear probability model** for $P(A_i = 1 \mid X_i)$, so $\tilde{A}_i = A_i - \hat{e}^{\text{lin}}(X_i)$. OLS carries a propensity model inside it: linear in X , fit by least squares.



OLS is DR? Step 2: two models inside one regression

One regression, two working models: **outcome** $E[Y | A, X] = \tau A + X^\top \beta$ and **implicit propensity** $E[A | X] = X^\top \gamma$.

OLS is DR? Step 2: two models inside one regression

One regression, two working models: **outcome** $E[Y | A, X] = \tau A + X^\top \beta$ and **implicit propensity** $E[A | X] = X^\top \gamma$.

Route 1: outcome model right. $Y_i = \tau A_i + X_i^\top \beta + \varepsilon_i$ with $E[\varepsilon_i | A_i, X_i] = 0$, so $\hat{\tau}_{OLS} \rightarrow \tau$. Textbook.



OLS is DR? Step 2: two models inside one regression

One regression, two working models: **outcome** $E[Y | A, X] = \tau A + X^\top \beta$ and **implicit propensity** $E[A | X] = X^\top \gamma$.

Route 1: outcome model right. $Y_i = \tau A_i + X_i^\top \beta + \varepsilon_i$ with $E[\varepsilon_i | A_i, X_i] = 0$, so $\hat{\tau}_{OLS} \rightarrow \tau$. Textbook.

Route 2: propensity model right, outcome model wrong. Let $Y_i = \tau A_i + g(X_i) + \varepsilon_i$ with g arbitrary. Substitute into FWL, using $\sum_i \tilde{A}_i A_i = \sum_i \tilde{A}_i^2$:

$$\hat{\tau}_{OLS} - \tau = \frac{\sum_i \tilde{A}_i g(X_i) + \sum_i \tilde{A}_i \varepsilon_i}{\sum_i \tilde{A}_i^2}$$

If $E[A | X] = X^\top \gamma$, then in the population $\tilde{A} = A - X^\top \gamma$ has $E[\tilde{A} | X] = 0$, so by iterated expectations $E[\tilde{A} g(X)] = E[E[\tilde{A} | X] g(X)] = 0$ for *every* g .



OLS is DR? Step 2: two models inside one regression

One regression, two working models: **outcome** $E[Y | A, X] = \tau A + X^\top \beta$ and **implicit propensity** $E[A | X] = X^\top \gamma$.

Route 1: outcome model right. $Y_i = \tau A_i + X_i^\top \beta + \varepsilon_i$ with $E[\varepsilon_i | A_i, X_i] = 0$, so $\hat{\tau}_{OLS} \rightarrow \tau$. Textbook.

Route 2: propensity model right, outcome model wrong. Let $Y_i = \tau A_i + g(X_i) + \varepsilon_i$ with g arbitrary. Substitute into FWL, using $\sum_i \tilde{A}_i A_i = \sum_i \tilde{A}_i^2$:

$$\hat{\tau}_{OLS} - \tau = \frac{\sum_i \tilde{A}_i g(X_i) + \sum_i \tilde{A}_i \varepsilon_i}{\sum_i \tilde{A}_i^2}$$

If $E[A | X] = X^\top \gamma$, then in the population $\tilde{A} = A - X^\top \gamma$ has $E[\tilde{A} | X] = 0$, so by iterated expectations $E[\tilde{A} g(X)] = E[E[\tilde{A} | X] g(X)] = 0$ for *every* g .

- › The omitted nonlinearity is **orthogonal to the residualised treatment**, so $\hat{\tau}_{OLS} \rightarrow \tau$ with $E[Y | A, X]$ misspecified. Same structure as Bang & Robins Step 2.



OLS is DR? Step 3: the heteroskedasticity catch

Route 2 assumed a constant τ . Let effects vary, $\tau(X) = E[Y(1) - Y(0) \mid X]$, with only the propensity model right, so $\tilde{A} = A - e(X)$. Conditional on X :

$$E[\tilde{A} Y \mid X] = e(X)(1 - e(X)) \tau(X), \quad E[\tilde{A}^2 \mid X] = e(X)(1 - e(X))$$



OLS is DR? Step 3: the heteroskedasticity catch

Route 2 assumed a constant τ . Let effects vary, $\tau(X) = E[Y(1) - Y(0) \mid X]$, with only the propensity model right, so $\tilde{A} = A - e(X)$. Conditional on X :

$$E[\tilde{A} Y \mid X] = e(X)(1 - e(X)) \tau(X), \quad E[\tilde{A}^2 \mid X] = e(X)(1 - e(X))$$

$$\hat{\tau}_{\text{OLS}} \rightarrow \frac{E[\tilde{A} Y]}{E[\tilde{A}^2]} = \frac{E[e(X)(1 - e(X)) \tau(X)]}{E[e(X)(1 - e(X))]}$$



OLS is DR? Step 3: the heteroskedasticity catch

Route 2 assumed a constant τ . Let effects vary, $\tau(X) = E[Y(1) - Y(0) \mid X]$, with only the propensity model right, so $\tilde{A} = A - e(X)$. Conditional on X :

$$E[\tilde{A} Y \mid X] = e(X)(1 - e(X)) \tau(X), \quad E[\tilde{A}^2 \mid X] = e(X)(1 - e(X))$$

$$\hat{\tau}_{\text{OLS}} \rightarrow \frac{E[\tilde{A} Y]}{E[\tilde{A}^2]} = \frac{E[e(X)(1 - e(X)) \tau(X)]}{E[e(X)(1 - e(X))]}$$

- › This is ([Aronow and Samii 2016](#)): a **conditional-variance-weighted** effect.
 - » Units with $e(X)$ near $1/2$ count most.
 - » The heteroskedasticity is in the **treatment**: $\text{Var}(A \mid X) = e(1 - e)$.
- › It equals the ATE when $\tau(X)$ is constant or uncorrelated with $e(X)(1 - e(X))$.
 - » Otherwise OLS is doubly robust for a *different estimand*.



OLS is DR? Step 3: the heteroskedasticity catch

Route 2 assumed a constant τ . Let effects vary, $\tau(X) = E[Y(1) - Y(0) | X]$, with only the propensity model right, so $\tilde{A} = A - e(X)$. Conditional on X :

$$E[\tilde{A} Y | X] = e(X)(1 - e(X)) \tau(X), \quad E[\tilde{A}^2 | X] = e(X)(1 - e(X))$$

$$\hat{\tau}_{\text{OLS}} \rightarrow \frac{E[\tilde{A} Y]}{E[\tilde{A}^2]} = \frac{E[e(X)(1 - e(X)) \tau(X)]}{E[e(X)(1 - e(X))]}$$

- › This is ([Aronow and Samii 2016](#)): a **conditional-variance-weighted** effect.
 - » Units with $e(X)$ near $1/2$ count most.
 - » The heteroskedasticity is in the **treatment**: $\text{Var}(A | X) = e(1 - e)$.
- › It equals the ATE when $\tau(X)$ is constant or uncorrelated with $e(X)(1 - e(X))$.
 - » Otherwise OLS is doubly robust for a *different estimand*.

Punchline: OLS is DR for a constant effect because it balances X exactly in sample.

Bang & Robins put $\hat{e}^{-1}$ into the outcome regression on purpose; OLS puts a linear $\hat{e}$ in by accident.



Bang & Robins simulation results

Setup: Multiple data-generating processes with different model specifications

1. When both models correct:

- › Low bias, good coverage
- › Achieves efficiency gains over single-model approaches

2. When only outcome model correct:

- › $\hat{\mu}_{DR}$ remains consistent
- › Outperforms pure IPW approaches ($\hat{\mu}_{HT}$)

3. When only propensity score correct:

- › $\hat{\mu}_{DR}$ remains consistent
- › Outperforms pure outcome regression ($\hat{\mu}_{OR}$)

4. When both models wrong: All methods can fail.

The “double protection”: DR gives you robustness against misspecification of either model, dramatically improving practical performance.

Table 2
Simulation result: estimating $\mu = E(Y)$ (upper) and treatment effect (lower)

Estimator	Bias ¹	Variance	Bias ²	IQR
$\hat{\mu}_{HT}$	-0.01	0.11	0.00	0.43
$\hat{\mu}_{HT, \text{fal}}$	-0.31	0.12	-0.29	0.44
$\hat{\mu}_{OR}$	0.00	0.04	-0.00	0.28
$\hat{\mu}_{OR, \text{fal}}$	-0.36	0.12	-0.35	0.44
$\hat{\mu}_{dr}$	0.00	0.04	-0.00	0.28
$\hat{\mu}_{dr, \text{ofal}}$	-0.02	0.11	0.00	0.45
$\hat{\mu}_{dr, \text{pfal}}$	0.00	0.04	-0.00	0.28
$\hat{\mu}_{dr, \text{ofal}} \oplus \text{pfal}$	-0.32	0.12	-0.30	0.45
$\hat{\mu}_{HT}$	-0.01	0.21	-0.00	0.59
$\hat{\mu}_{HT, \text{fal}}$	0.86	0.15	0.87	0.52
$\hat{\mu}_{OR}$	0.00	0.07	0.01	0.35
$\hat{\mu}_{OR, \text{fal}}$	-1.56	0.07	-1.56	0.34
$\hat{\mu}_{dr}$	0.00	0.09	0.01	0.41
$\hat{\mu}_{dr, \text{ofal}}$	-0.09	0.28	-0.06	0.63
$\hat{\mu}_{dr, \text{pfal}}$	0.00	0.08	0.01	0.39
$\hat{\mu}_{dr, \text{ofal}} \oplus \text{pfal}$	0.92	0.15	0.93	0.54

True parameter μ is 1 for mean parameter (upper) and 2 for treatment effect (lower). Bias¹, Bias², and Variance denote bias in mean, bias in median, and variance of the estimates from 1000 simulations, respectively. IQR denotes the interquartile range, that is, upper quartile (75%)–lower quartile (25%). Each simulation is based on the sample size of 500.

What Bang & Robins opened up

Insight: We don't need to get *one* model exactly right. We can hedge our bets.

Key innovations:

- › Unified framework connecting outcome regression and IPW
- › Formal double robustness theory
- › Practical estimators that “just work” in many settings

Issues:

- › Why should we expect to get even one model right?
 - » Yes, two lottery tickets give you a better chance than one.
- › How should we do all of this estimation flexibly?
 - » Replace parametric models with ML, IPW with balancing weights?

Open questions

1. **Machine learning integration:** What happens when we soup-up Frisch-Waugh-Lovell with machine learning?
 - › *Motivates:* ([Chernozhukov et al. 2018](#)) on double/debiased ML
2. **Optimal bias-variance tradeoffs:** Can we do better than simple plug-in estimation? How do we target the parameter of interest more directly?
 - › *Motivates:* ([Schuler and Rose 2017](#)) on targeted maximum likelihood estimation (TMLE), with more detail in ([Van der Laan, Rose, et al. 2011](#)).
3. **When do DR methods actually work?** Are there settings where all approaches fail catastrophically?
 - › *Motivates:* ([Kang and Schafer 2007](#)) empirical comparison studies
4. **Modern theoretical foundations:** How do we unify these ideas with modern semiparametric theory and machine learning?
 - › *Motivates:* ([Kennedy 2022](#)) comprehensive review



More to read

Paper	Why you'd read it
(Kennedy 2022)	The review that frames today: influence functions, and why “doubly robust” and “debiased” are one idea. Parts 1–3.
(Chernozhukov et al. 2018)	Double/debiased ML: Neyman-orthogonal scores plus cross-fitting make ML nuisance estimates safe to plug in. Parts 1–3.
(Schuler and Rose 2017) ; (Van der Laan, Rose, et al. 2011)	TMLE: fit an initial outcome model, then take one <i>targeting</i> step toward the parameter. The tutorial, and the book behind it (Ch. 4–6).
(Bruns-Smith et al. 2025)	Augmented balancing weights are a linear regression in disguise: last week's methods and this week's are one estimator.
(Chernozhukov et al. 2022) ; (Chernozhukov, Newey, and Singh 2022)	Automatic debiased ML: learn the Riesz representer directly, by neural net or forest, instead of a propensity score. In EconML .
(Kang and Schafer 2007)	The cautionary tale: every DR estimator does badly when both models are mildly wrong.



References

- Aronow, P M, and Cyrus Samii. 2016. "Does Regression Produce Representative Estimates of Causal Effects?" *American Journal of Political Science* 60 (1): 250–67.
- Bang, Heejung, and James M Robins. 2005. "Doubly Robust Estimation in Missing Data and Causal Inference Models." *Biometrics* 61 (4): 962–73.
- Bruns-Smith, David, Oliver Dukes, Avi Feller, and Elizabeth L Ogburn. 2025. "Augmented Balancing Weights as Linear Regression." *Journal of the Royal Statistical Society: Series B* 88 (3): 699–723.
- Chernozhukov, Victor, Denis Chetverikov, Mert Demirer, Esther Duflo, Christian Hansen, Whitney Newey, and James Robins. 2018. "Double/Debiased Machine Learning for Treatment and Structural Parameters." *The Econometrics Journal* 21 (1): C1–68. <https://doi.org/10.1111/ectj.12097>.
- Chernozhukov, Victor, Whitney K Newey, Víctor Quintas-Martínez, and Vasilis Syrgkanis. 2022. "RieszNet and ForestRiesz: Automatic Debiased Machine Learning with Neural Nets and Random Forests." In *International Conference on Machine Learning*, 3901–14. PMLR.
- Chernozhukov, Victor, Whitney K Newey, and Rahul Singh. 2022. "Automatic Debiased Machine Learning of Causal and Structural Effects." *Econometrica* 90 (3): 967–1027.
- Kang, Joseph DY, and Joseph L Schafer. 2007. "Demystifying Double Robustness: A Comparison of Alternative Strategies for Estimating a Population Mean from Incomplete Data." *Statistical Science* 22 (4): 523–39.
- Kennedy, Edward H. 2022. "Semiparametric Doubly Robust Targeted Double Machine Learning: A Review." <https://arxiv.org/abs/2203.06469>.
- Schuler, Megan S., and Sherri Rose. 2017. "Targeted Maximum Likelihood Estimation for Causal Inference in Observational Studies." *American Journal of Epidemiology* 185 (1): 65–73. <https://doi.org/10.1093/aje/kww165>.
- Van der Laan, Mark J, Sherri Rose, et al. 2011. *Targeted Learning: Causal Inference for Observational and Experimental Data*. Vol. 4. Springer.

